UNIVERSITÉ CLAUDE BERNARD LYON 1

Master 2 – DISS

Data Processing and Analytics

Analyzing New York City Yellow Taxi Demand

Volume Analysis Using Apache Spark SQL

Project Part 1 – Volume

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Submission date: 29 October 2026

1 Research Question

This project studies how yellow taxi activity varies across New York City. The overall objective is to understand how trips are distributed over time and across taxi zones, and how these zones are connected through journeys.

For Part 1 (Volume), the analysis focuses on the following sub-question:

How does yellow taxi pickup demand vary by taxi zone, hour of the day, and day of the week?

The objective is to process a large volume of trip records and identify the main spatial and temporal patterns in pickup demand.

2 Data Sources

Two datasets published by the New York City Taxi and Limousine Commission (TLC) were used.

- **Yellow Taxi Trip Records (January 2022 – December 2024):** monthly Parquet files containing individual taxi trips, including pickup and drop-off timestamps, location identifiers, trip distance, and payment information.
- **Taxi Zone Lookup Table:** a CSV file mapping location identifiers to zone names, boroughs, and service zones.

The trip records were downloaded from the official TLC data distribution service. The lookup table was obtained from the same source.

Official source: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

The data is publicly accessible. Users should consult the TLC website for the applicable data-use terms and any updates. The TLC also notes that the records are provided by technology providers and may contain inaccuracies.

The acquisition script downloads all 36 monthly trip files and the lookup table. The downloaded raw files represent approximately **1.81 GB** of data.

3 Technology and Processing Choice

Apache Spark was selected because the dataset contains more than 119 million trip records. Spark supports distributed data processing, SQL-based aggregation, and Parquet storage.

The pipeline was implemented using PySpark 3.5.8 and Python 3.11. The processing was executed locally through Docker.

The source files were read using explicit schemas and stored as typed Parquet datasets before cleaning and analysis.

GitHub repository: https://github.com/miaa-T/nyc-taxi-volume

4 Data Processing Pipeline

The processing pipeline consists of the following stages:

1. **Acquisition:** download the 36 monthly Yellow Taxi files and the taxi zone lookup table.
2. **Ingestion:** read the source data with explicit Spark schemas and store the records in Parquet format, partitioned by year and month.
3. **Cleaning:** remove trips whose drop-off timestamp is earlier than their pickup timestamp.
4. **Joining:** enrich each trip with its pickup zone name, borough, and service zone using the lookup table.
5. **Analysis:** use Spark SQL to calculate pickup counts by zone, hour, and day of the week.
6. **Validation and visualisation:** check aggregate totals and generate charts from the analysis results.

5 Data Cleaning and Validation

The initial ingestion produced **119,136,044 trip records** covering the period from January 2022 to December 2024.

Data profiling identified 17,663 records where the drop-off timestamp was earlier than the pickup timestamp. These records were removed because they represent inconsistent trip durations.

After cleaning, **119,118,381 records** remained.

Cleaning indicator	Number of records
Initial records	119,136,044
Invalid date records removed	17,663
Records after cleaning	119,118,381

Table 1: Summary of the cleaning operation.

Other observations included missing passenger counts, zero trip distances, and negative fare or total amounts. These values were not automatically removed because they do not directly affect the pickup demand analysis and may correspond to special cases or adjustments.

A duplicate check on January 2022 found no exact duplicate rows when all columns were considered. Records sharing a subset of attributes were retained because they cannot necessarily be considered duplicate trips.

The cleaned trip dataset was joined with the taxi zone lookup table using `PULocationID = LocationID`. The lookup contained **265 taxi zones**. The join preserved all **119,118,381 trips**, indicating that every pickup location in the cleaned dataset matched a lookup entry.

6 Analysis Method

Spark SQL was used to calculate the number of pickups for each:

- pickup zone and borough;
- hour of the day (0–23);
- day of the week.

The three aggregations were validated by comparing their total pickup counts with the number of records in the joined dataset. Each aggregation returned **119,118,381 pickups**, confirming that the analysis covered the complete cleaned dataset.

7 Results and Discussion

7.1 Spatial Distribution of Pickup Demand

The highest pickup count was recorded at **JFK Airport**, with 5,897,623 pickups over the three-year period. Several Manhattan zones also appear among the most active areas, including Upper East Side South (5,566,198 pickups) and Midtown Center (5,253,617 pickups).

Both JFK Airport and LaGuardia Airport appear in the top ten zones. This suggests that airport areas and central Manhattan locations account for a substantial share of the observed pickup activity.

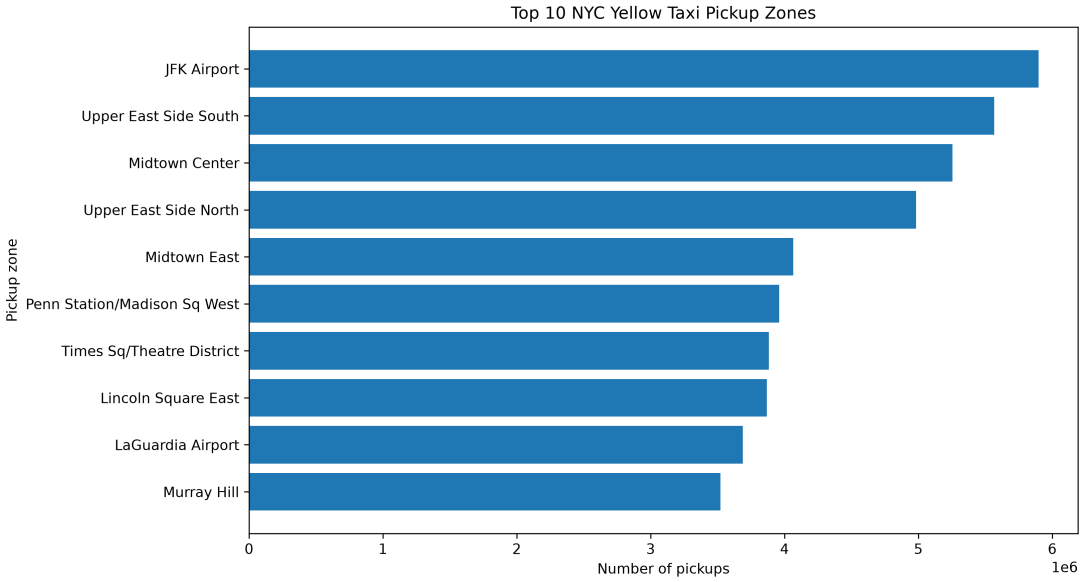


Figure 1: Ten taxi zones with the highest pickup counts.

7.2 Hourly Distribution

Pickup demand changes considerably throughout the day. The lowest number of pickups occurs at 04:00, with 681,496 trips. Activity increases during the morning and reaches its maximum at 18:00, with 8,457,664 pickups.

The period between 17:00 and 19:00 is particularly active, indicating a strong concentration of trips in the late afternoon and early evening.

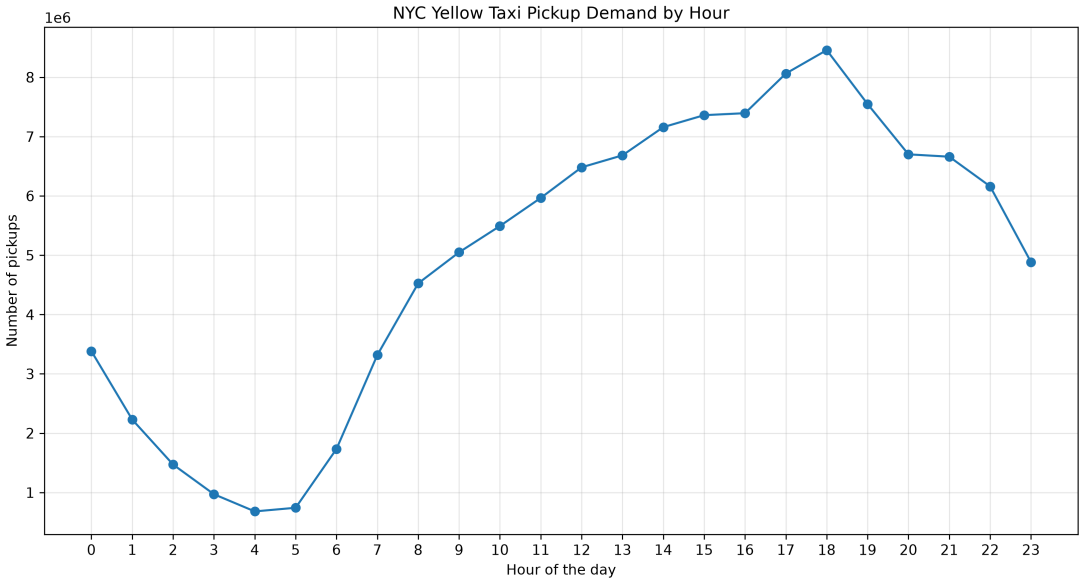


Figure 2: Pickup demand by hour of the day.

7.3 Weekly Distribution

Thursday is the busiest day, with 18,561,008 pickups, while Sunday has the lowest count, with 14,899,572 pickups. Tuesday, Wednesday, Friday, and Saturday also show relatively high activity compared with Sunday and Monday.

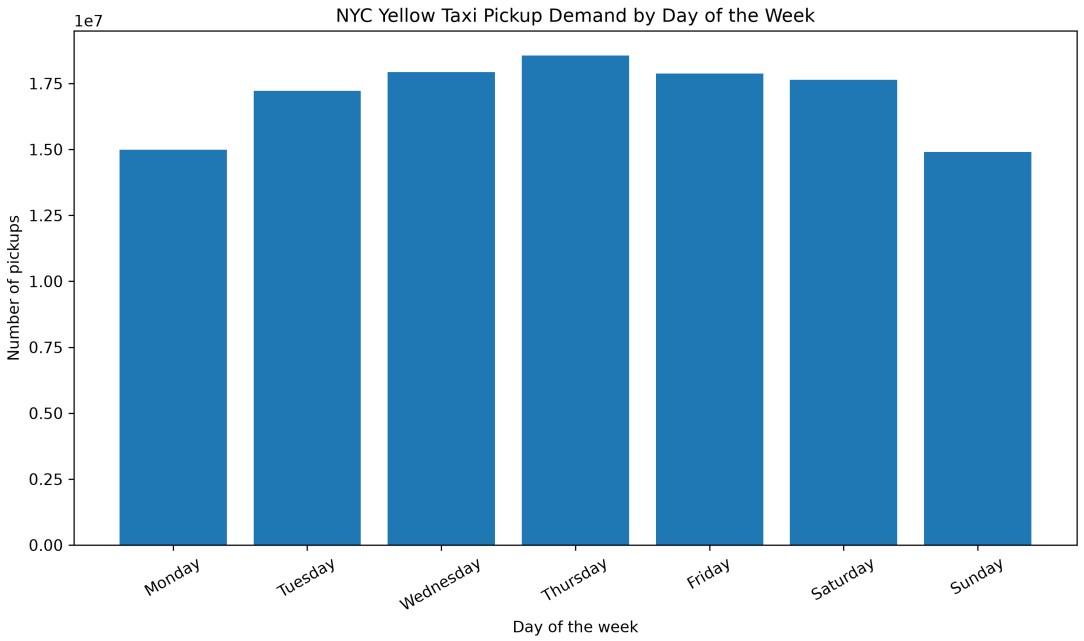


Figure 3: Pickup demand by day of the week.

8 Conclusion

The analysis shows that yellow taxi pickup demand is not evenly distributed across New York City or throughout the week. Airport zones and central Manhattan locations record particularly high pickup volumes. At the hourly level, demand reaches its maximum at 18:00, while Thursday is the busiest day in the analysed period.

The results answer the Part 1 sub-question by identifying clear spatial and temporal differences in pickup activity. The pipeline also demonstrates how Apache Spark can process and aggregate a large dataset while preserving reproducibility through scripted acquisition, typed Parquet storage, cleaning, joining, and validation.